

Collector Deployment Patterns

Series: OTEL | Notebook: 3 of 8 | Created: January 2026

Deploying the OpenTelemetry Collector

The OTel Collector can be deployed in various patterns depending on your infrastructure. This notebook covers deployment modes, Kubernetes configurations, and best practices for production.

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Prerequisites

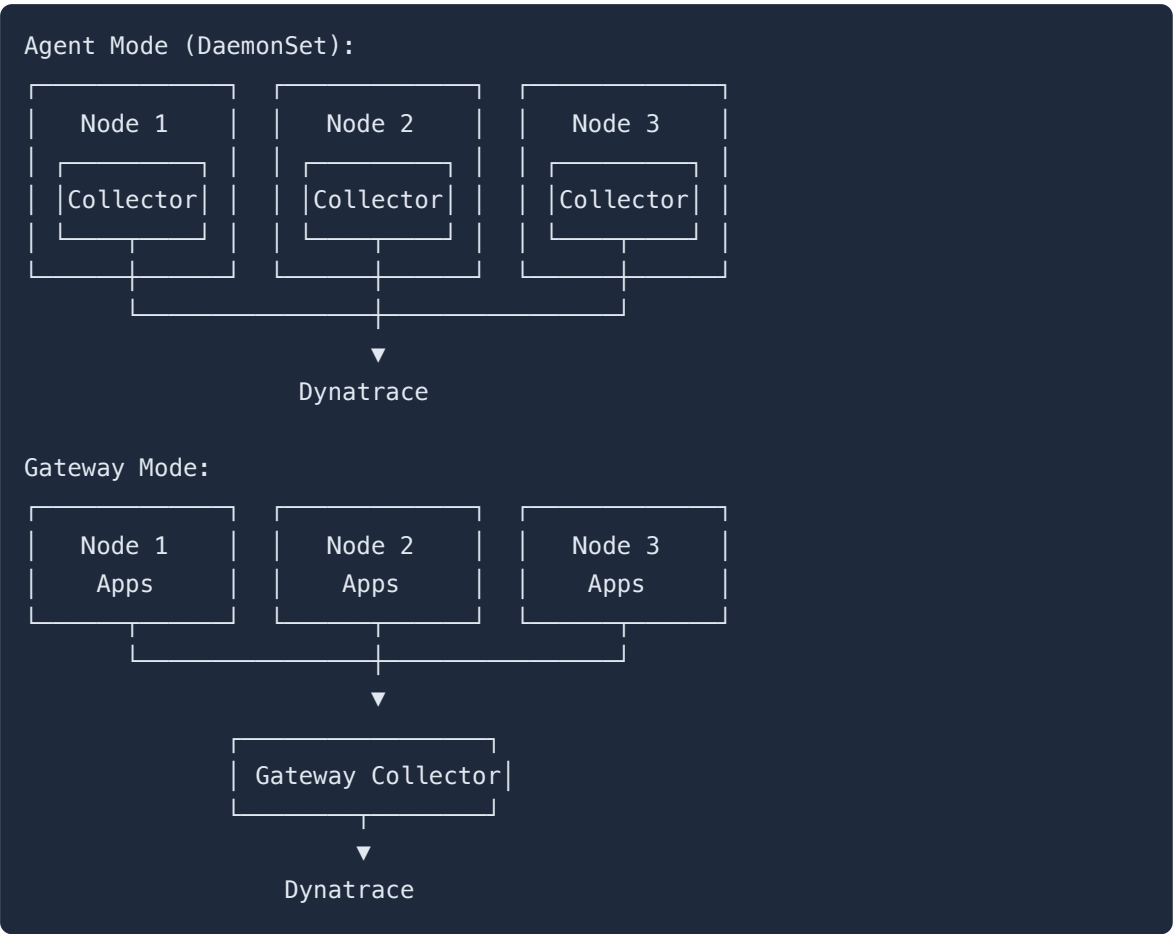
Requirement	Details
Knowledge	OTEL-02: Collector Architecture
Tools	kubect!, Helm, or Docker

1. Deployment Modes

Mode Comparison

Mode	Deployment	Use Case	Pros	Cons
Agent	DaemonSet/Sidecar	Per-node collection	Low latency, local processing	More instances
Gateway	Deployment	Centralized	Single point, aggregation	Potential bottleneck
Hybrid	Both	Large scale	Best of both	Complex

Architecture Patterns



2. Agent Mode (Sidecar/DaemonSet)

DaemonSet Configuration

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: otel-collector-agent
  namespace: otel
spec:
  selector:
    matchLabels:
      app: otel-collector-agent
  template:
    metadata:
      labels:
        app: otel-collector-agent
    spec:
      containers:
        - name: collector
          image: otel/opentelemetry-collector-contrib:latest
          args:
            - --config=/conf/otel-collector-config.yaml
          ports:
            - containerPort: 4317 # OTLP gRPC
            - containerPort: 4318 # OTLP HTTP
          resources:
            requests:
              cpu: 100m
              memory: 256Mi
            limits:
              cpu: 500m
              memory: 512Mi
          volumeMounts:
            - name: config
              mountPath: /conf
      volumes:
        - name: config
          configMap:
            name: otel-collector-config
```

Sidecar Pattern

```
apiVersion: v1
kind: Pod
metadata:
  name: my-app
spec:
  containers:
    - name: app
      image: my-app:latest
      env:
        - name: OTEL_EXPORTER_OTLP_ENDPOINT
          value: http://localhost:4317
    - name: otel-collector
      image: otel/opentelemetry-collector-contrib:latest
      args:
        - --config=/conf/otel-collector-config.yaml
      ports:
        - containerPort: 4317
```

3. Gateway Mode

Gateway Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: otel-collector-gateway
  namespace: otel
spec:
  replicas: 2 # HA
  selector:
    matchLabels:
      app: otel-collector-gateway
  template:
    metadata:
      labels:
        app: otel-collector-gateway
    spec:
      containers:
        - name: collector
          image: otel/opentelemetry-collector-contrib:latest
          args:
            - --config=/conf/otel-collector-config.yaml
          ports:
            - containerPort: 4317
            - containerPort: 4318
          resources:
            requests:
              cpu: 500m
              memory: 1Gi
            limits:
              cpu: 2000m
              memory: 4Gi
```

Gateway Service

```
apiVersion: v1
kind: Service
metadata:
  name: otel-collector-gateway
  namespace: otel
spec:
  ports:
    - name: otlp-grpc
      port: 4317
      targetPort: 4317
    - name: otlp-http
      port: 4318
      targetPort: 4318
  selector:
    app: otel-collector-gateway
```

4. Kubernetes Deployment

Helm Installation

```
# Add Helm repo
helm repo add open-telemetry https://open-telemetry.github.io/opentelemetry-
helm-charts
helm repo update

# Install as DaemonSet (agent mode)
helm install otel-collector open-telemetry/opentelemetry-collector \
  --namespace otel \
  --create-namespace \
  --set mode=daemonset

# Install as Deployment (gateway mode)
helm install otel-collector open-telemetry/opentelemetry-collector \
  --namespace otel \
  --create-namespace \
  --set mode=deployment \
  --set replicaCount=2
```

Helm Values for Dynatrace

```
# values.yaml
mode: deployment
replicaCount: 2

config:
  receivers:
    otlp:
      protocols:
        grpc:
          endpoint: 0.0.0.0:4317
        http:
          endpoint: 0.0.0.0:4318

  processors:
    batch:
      timeout: 10s
    memory_limiter:
      check_interval: 1s
      limit_mib: 800

  exporters:
    otlphttp:
      endpoint: https://${DT_ENDPOINT}/api/v2/otlp
      headers:
        Authorization: Api-Token ${DT_TOKEN}

  service:
    pipelines:
      traces:
        receivers: [otlp]
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
      metrics:
        receivers: [otlp]
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
      logs:
        receivers: [otlp]
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
```

5. Docker Compose

Basic Setup

```
# docker-compose.yaml
version: '3.8'

services:
  otel-collector:
    image: otel/opentelemetry-collector-contrib:latest
    command: ["--config=/etc/otel-collector-config.yaml"]
    volumes:
      - ./otel-collector-config.yaml:/etc/otel-collector-config.yaml
    ports:
      - "4317:4317" # OTLP gRPC
      - "4318:4318" # OTLP HTTP
      - "13133:13133" # Health check
    environment:
      - DT_ENDPOINT=${DT_ENDPOINT}
      - DT_TOKEN=${DT_TOKEN}

  my-app:
    image: my-app:latest
    environment:
      - OTEL_EXPORTER_OTLP_ENDPOINT=http://otel-collector:4317
      - OTEL_SERVICE_NAME=my-app
    depends_on:
      - otel-collector
```

6. High Availability

HA Considerations

Component	HA Strategy
Agent (DaemonSet)	Node failure = pod restarts
Gateway	Multiple replicas behind load balancer
Persistent Queue	Prevent data loss during restarts

Gateway with Persistent Queue

```
exporters:
  otlphttp:
    endpoint: https://${DT_ENDPOINT}/api/v2/otlp
    sending_queue:
      enabled: true
      num_consumers: 10
      queue_size: 1000
      storage: file_storage
    retry_on_failure:
      enabled: true
      initial_interval: 1s
      max_interval: 30s
      max_elapsed_time: 300s

extensions:
  file_storage:
    directory: /var/lib/otel/storage
    timeout: 10s

service:
  extensions: [file_storage]
```

Pod Disruption Budget

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: otel-collector-pdb
spec:
  minAvailable: 1
  selector:
    matchLabels:
      app: otel-collector-gateway
```

7. Resource Sizing

Sizing Guidelines

Throughput	CPU	Memory	Replicas
Low (<1000 spans/s)	100m	256Mi	1
Medium (<10k spans/s)	500m	1Gi	2
High (<100k spans/s)	2	4Gi	3+
Very High (>100k spans/s)	4+	8Gi+	5+

Memory Limiter Tuning

```
processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 800      # 80% of container memory limit
    spike_limit_mib: 200 # Room for spikes
```

Horizontal Pod Autoscaler

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: otel-collector-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: otel-collector-gateway
  minReplicas: 2
  maxReplicas: 10
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70
```

8. Security Considerations

Token Management

```
# Kubernetes Secret
apiVersion: v1
kind: Secret
metadata:
  name: dynatrace-otel-token
type: Opaque
stringData:
  token: dt0c01.xxx...
```

```
# Reference in Deployment
env:
  - name: DT_TOKEN
    valueFrom:
      secretKeyRef:
        name: dynatrace-otel-token
        key: token
```

TLS Configuration

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
        tls:
          cert_file: /certs/server.crt
          key_file: /certs/server.key
          client_ca_file: /certs/ca.crt # mTLS
```

Network Policy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: otel-collector-policy
spec:
  podSelector:
    matchLabels:
      app: otel-collector-gateway
  policyTypes:
    - Ingress
    - Egress
  ingress:
    - from:
        - namespaceSelector: {}
      ports:
        - protocol: TCP
          port: 4317
        - protocol: TCP
          port: 4318
  egress:
    - to:
        - ipBlock:
            cidr: 0.0.0.0/0
      ports:
        - protocol: TCP
          port: 443
```

Summary

In this notebook, you learned:

- Deployment modes: Agent vs. Gateway vs. Hybrid
- DaemonSet and Sidecar configurations for agent mode
- Gateway deployment with services
- Kubernetes deployment via Helm
- Docker Compose setup
- High availability with persistent queues
- Resource sizing guidelines
- Security best practices

Next Steps

Next Notebook	Topic
OTEL-04: Trace Instrumentation	Instrumenting code
OTEL-07: Dynatrace Integration	Complete DT setup

References

- [Collector Deployment](#)
 - [Helm Chart](#)
 - [Docker Images](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.